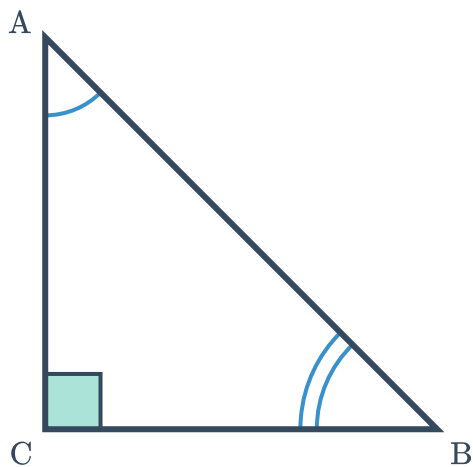


## Right-angled triangles

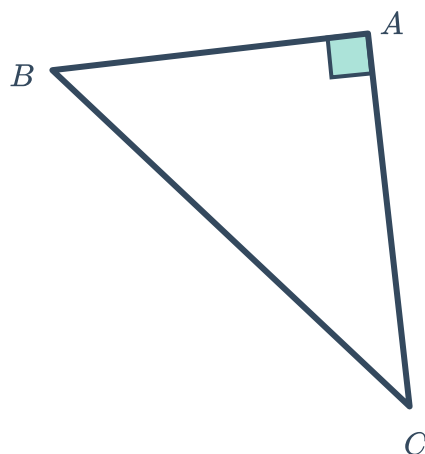


1 For the following triangles, name the hypotenuse:

a

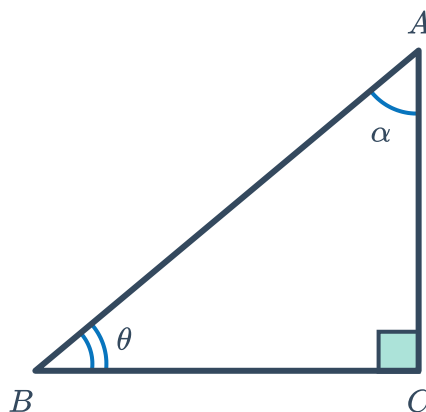


b



2 Consider the following triangle:

- a State the opposite side to angle  $\theta$ .
- b State the adjacent side to angle  $\theta$ .
- c State the opposite side to angle  $\alpha$ .
- d State the adjacent side to angle  $\alpha$ .
- e State the angle that is opposite the hypotenuse.



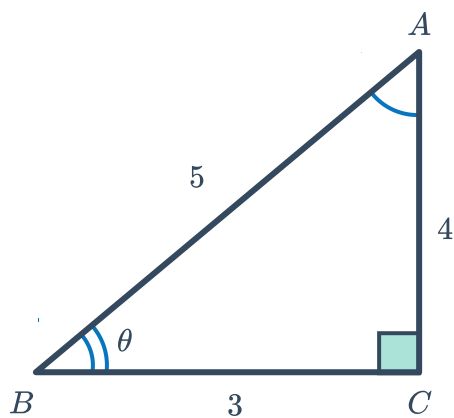
3 With reference the angle  $\theta$ , find the value of these ratios for each of the following triangles:

i  $\frac{\text{Opposite}}{\text{Adjacent}}$

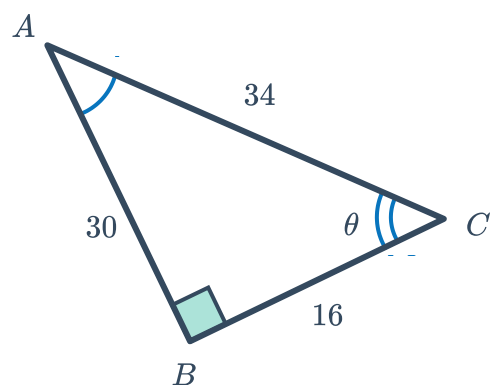
ii  $\frac{\text{Opposite}}{\text{Hypotenuse}}$

iii  $\frac{\text{Adjacent}}{\text{Hypotenuse}}$

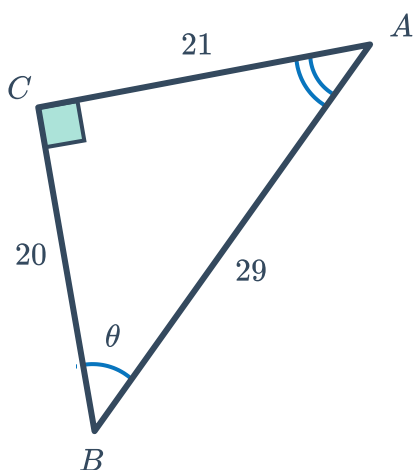
a



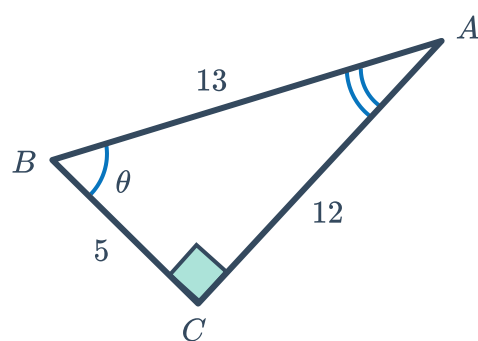
b



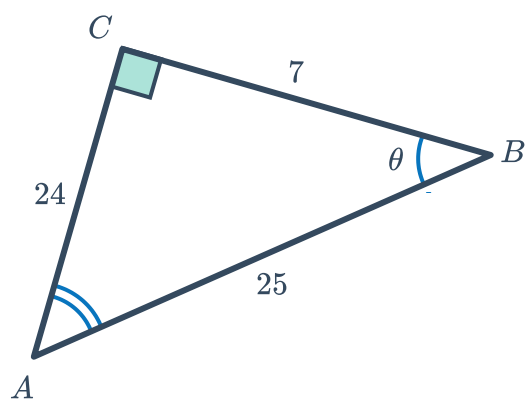
c



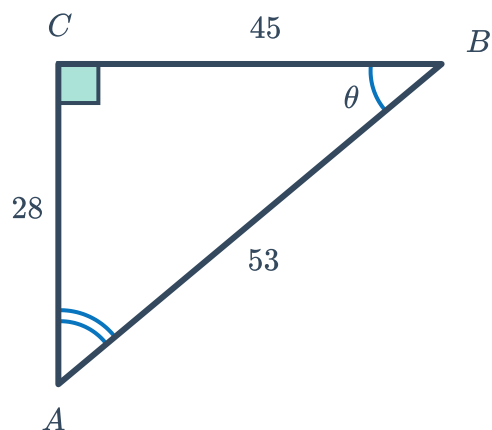
d



e



f





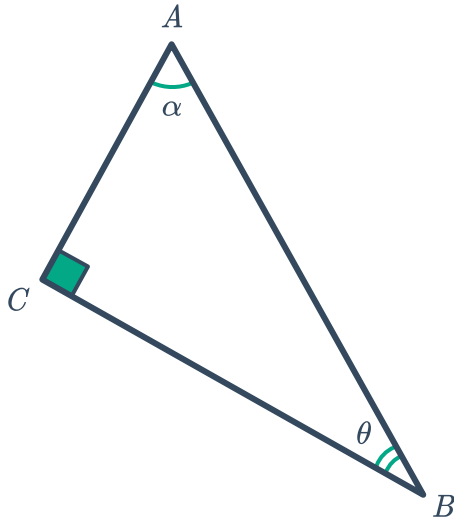
## Trigonometric ratios

4 Write down the indicated ratios for the following triangles:

a

i  $\tan \theta$

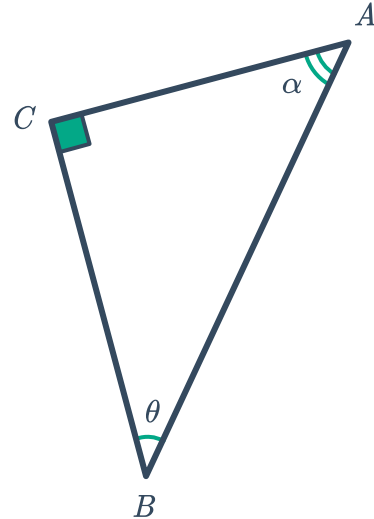
ii  $\sin \alpha$



b

i  $\sin \theta$

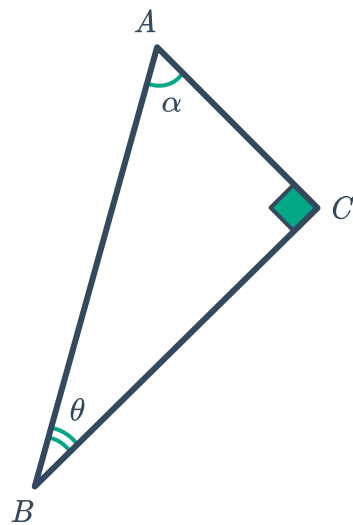
ii  $\tan \alpha$



c

i  $\cos \theta$

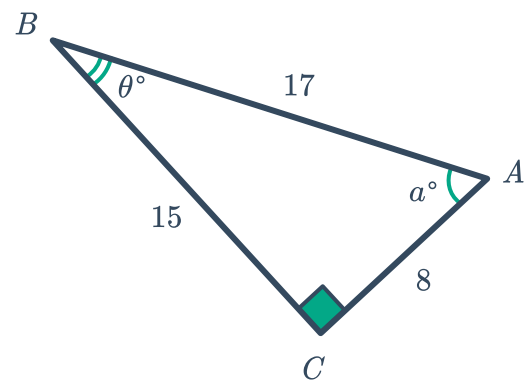
ii  $\cos \alpha$



d

i  $\tan \theta$

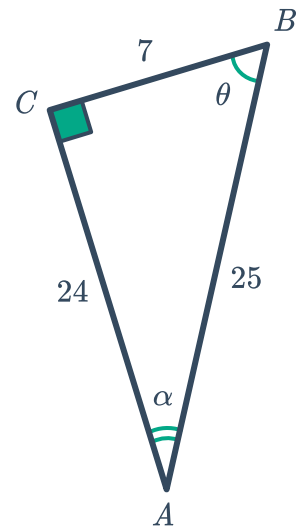
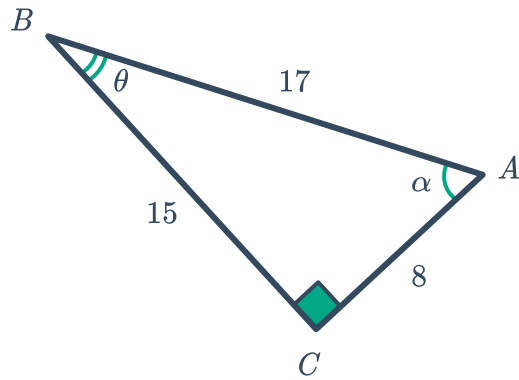
ii  $\sin \alpha$



e

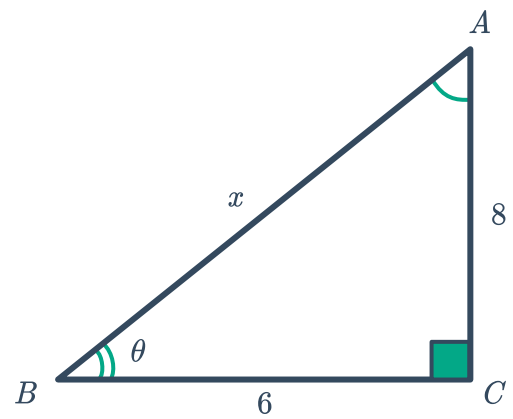
i  $\cos \theta$ ii  $\tan \alpha$ 

f

i  $\sin \theta$ ii  $\cos \alpha$ 

5 Consider the following triangle:

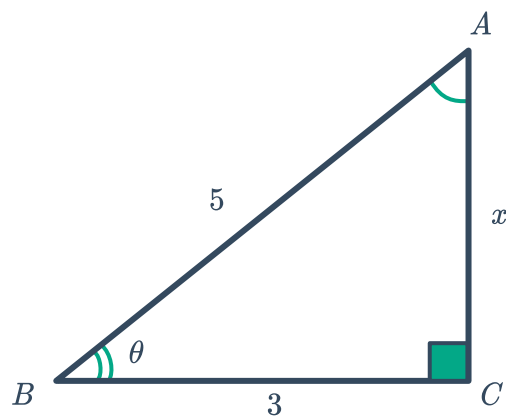
- a Find the value of  $x$ .
- b Find the value of  $\sin \theta$ .
- c Find the value of  $\cos \theta$ .



6 Consider the following triangle:

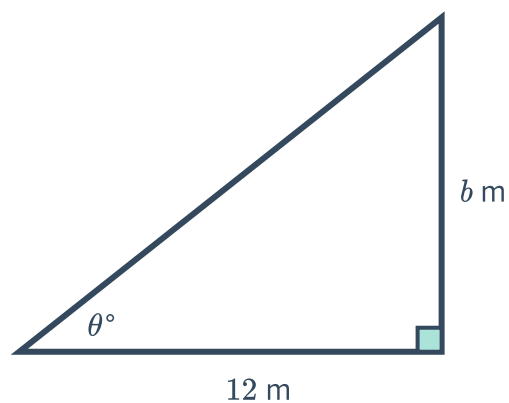


- a Find the value of  $x$ .
- b Hence, find the value of  $\tan \theta$ .

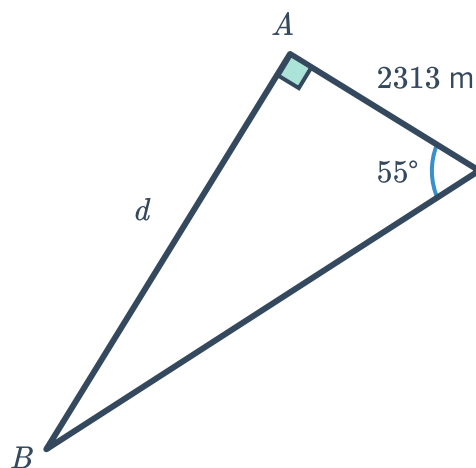


## Unknown sides

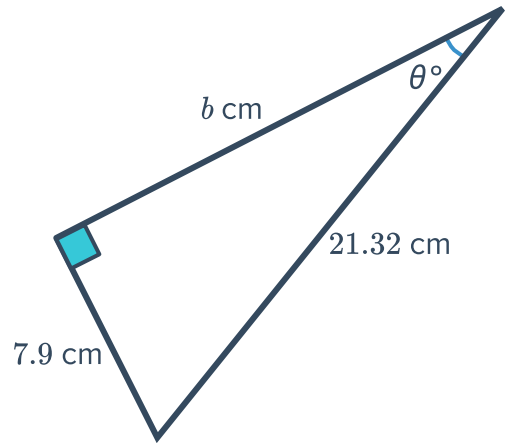
- 7 For the following triangle, if  $\tan \theta = \frac{2}{3}$ , find the value of  $b$ .



- 8 For the following triangle, if  $\tan \theta = \frac{4}{3}$ , find the value of  $d$ .

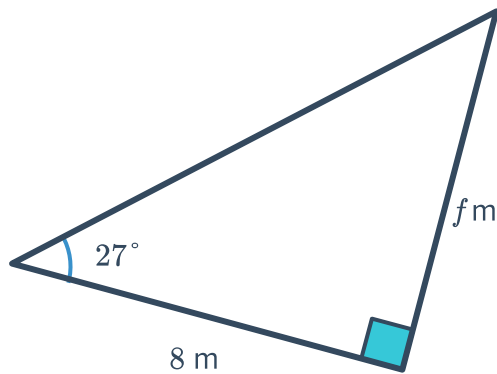


- 9 For the following triangle, if  $\tan \theta = 0.4$ , find the value of  $b$ .  
Round your answer correct to one decimal place.

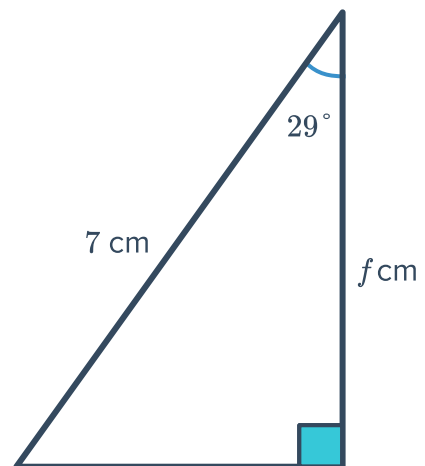


- 10 Find the value of the pronumeral in the following triangles, correct to two decimal places:

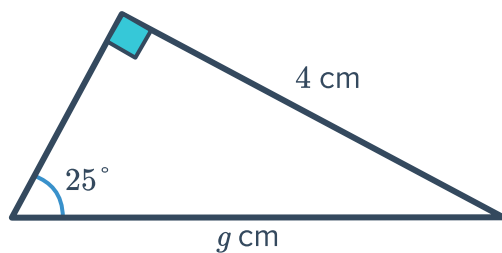
a



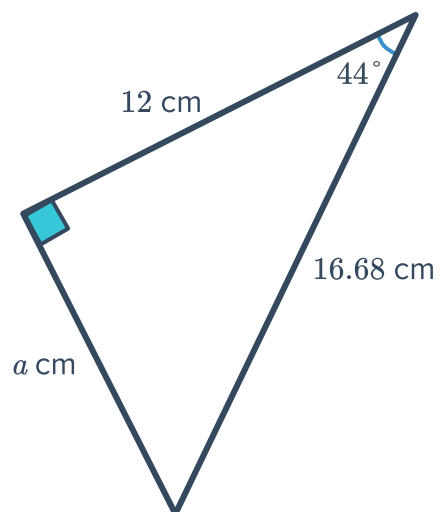
b



c



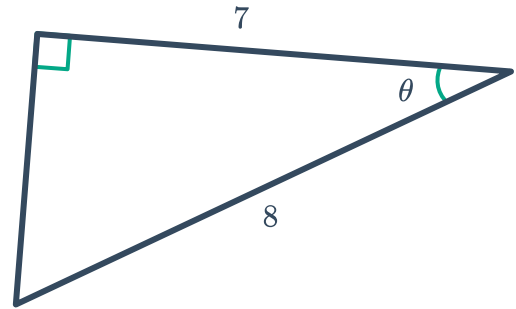
d





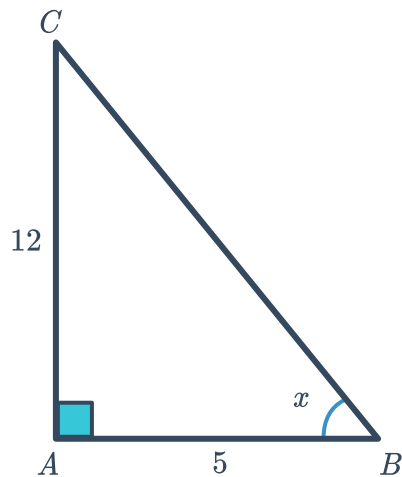
## Unknown angles

- 11 Given the following triangle, calculate the exact value of  $\tan \theta$ .

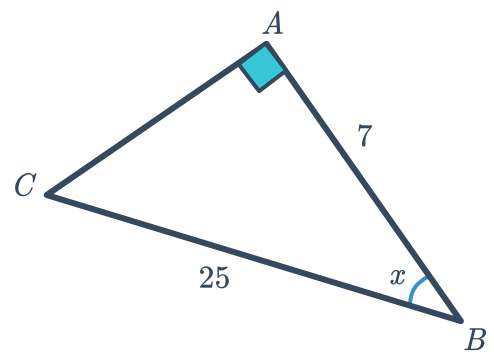


- 12 For each of the following triangles, find the value of  $x$  to the nearest degree:

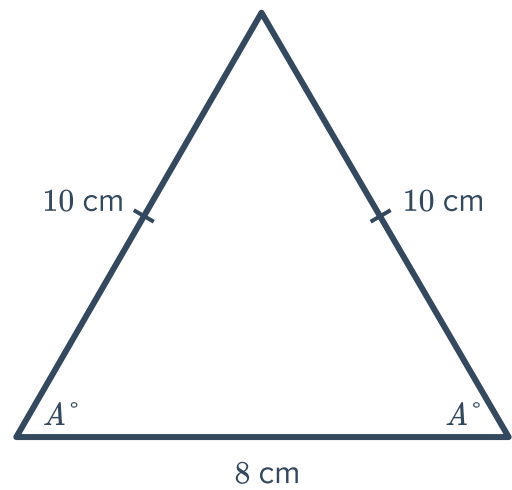
a



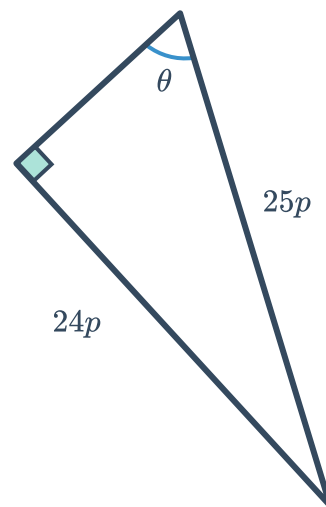
b



- 13 An isosceles triangle has equal side lengths of 10 cm and a base of 8 cm as shown. Calculate the size of angle  $A$  to one decimal place.



- 14 Find the value of  $\tan \theta$  for the following triangle:



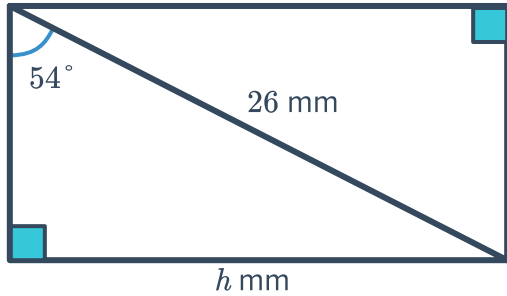




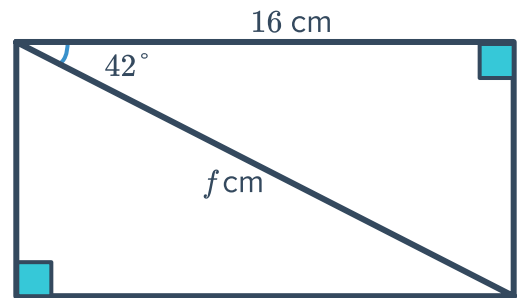
## Applications

15 Find the value of the pronumeral(s) in the following diagrams, correct to the nearest whole number:

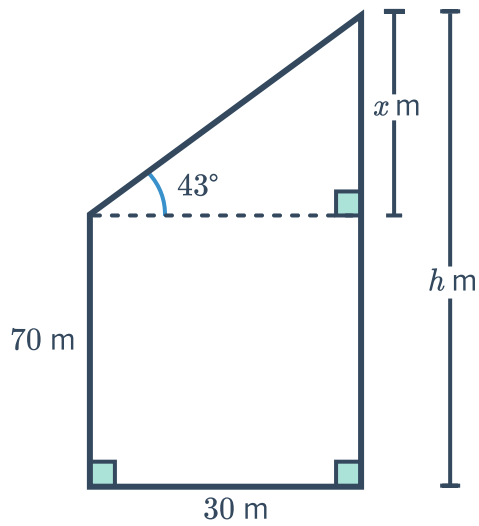
a



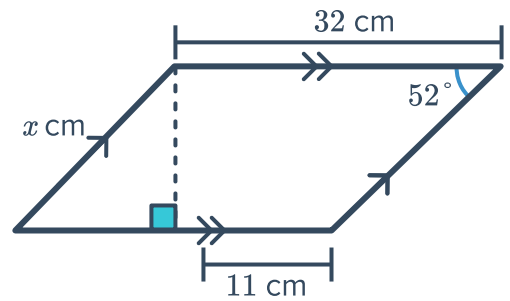
b



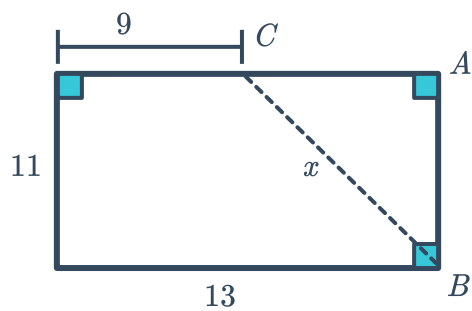
c



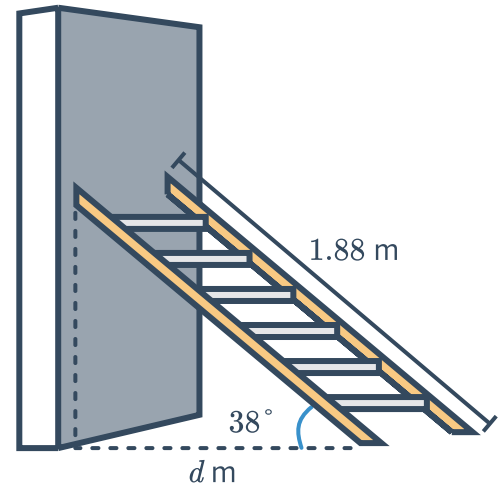
d



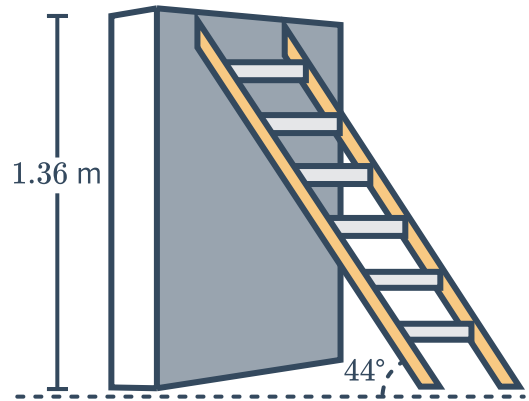
e



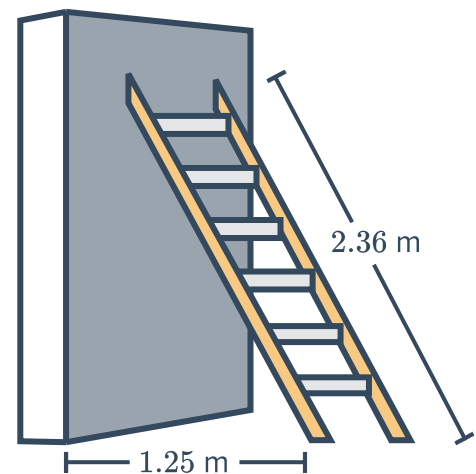
- 16 If  $d$  is the distance between the base of the wall and the base of the ladder, find the value of  $d$  to two decimal places.



- 17 A ladder is leaning at an angle of  $44^\circ$  against a 1.36 m high wall. Find the length of the ladder, to two decimal places.



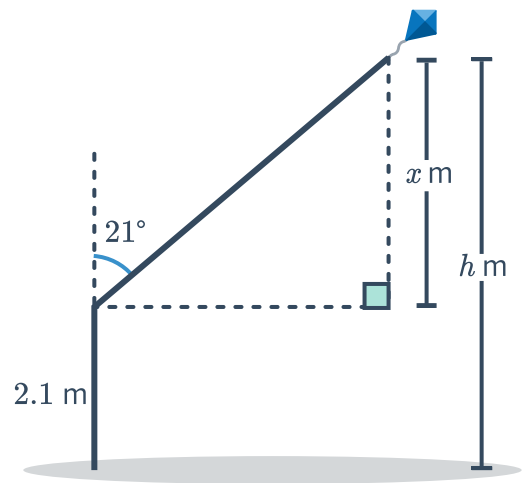
- 18 A ladder measuring 2.36 m in length is leaning against a wall. If the angle the ladder makes with the ground is  $y^\circ$ , find the value of  $y$  to two decimal places.



- 19 A girl is flying a kite that is attached to the end of a 23.4 m length of string. The angle between the string and the vertical is  $21^\circ$ . The girl is holding the string 2.1 m above the ground.



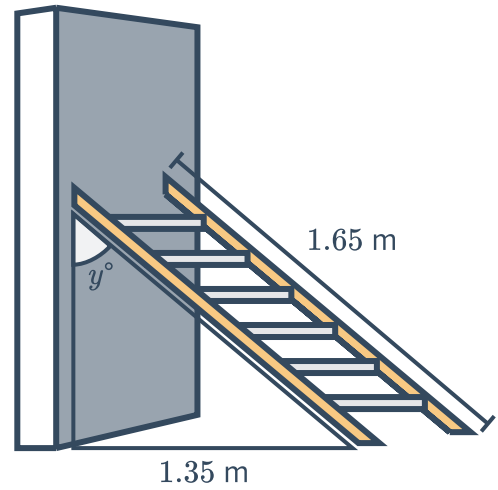
- a Find  $x$ , correct to two decimal places.
- b Hence, find the height,  $h$ , of the kite above the ground, correct to two decimal places.



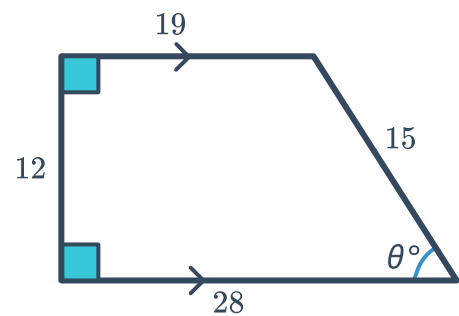
- 20 In the diagram, a string of lights joins the top of the tree to a point on the ground 23.9 m away. If the angle that the string of lights makes with the ground is  $\theta^\circ$ , find  $\theta$  to two decimal places.



- 21** A ladder measuring 1.65 m in length is leaning against a wall. If the angle the ladder makes with the wall is  $y^\circ$ , find  $y$  to two decimal places.



- 22** Find the value of  $\tan \theta$  in the following trapezium:



- 23** A sand pile has an angle of  $40^\circ$  and is 10.6 m wide. Find the height of the sand pile,  $h$ , to one decimal place.

